Cross-Lingual Correctness Probe Transfer Reveals Architecture-Dependent Pareto Frontiers: Discovery, Negative Result, and Practical Recommendation

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Abstract

Probe-based calibration methods exploit internal correctness directions to assess LLM confidence, but remain English-only. We present the first systematic evaluation of cross-lingual correctness probe transfer across 20 languages, two MoE architectures (Qwen3-235B-A22B, Llama 4 Scout), and one dense architecture (Gemma 4 31B), revealing three findings. **(1) Discovery:** A depth-transferability Pareto frontier emerges whose shape is architecture-dependent—Qwen3 shows a uniformly negative tradeoff (per-language Spearman $\rho \approx -0.50$) while Llama 4 Scout shows language-specific behavior (global $\rho = -0.12$; 4/20 languages exhibit *positive* ρ). **(2) Principled negative result:** CaCLR, a calibration-aware probe training objective combining BCE, smooth-binned ECE, and MMD consistency loss, fails to shift this frontier: a β sweep across three orders of magnitude yields negligible effect ($d = 0.07$, $p = 0.38$). CKA analysis suggests MMD matches marginals but cannot align the low-dimensional correctness subspace that rotates with depth. **(3) Practical recommendation:** Simple linear projection with $\sim$50 parallel pairs provides +3.5pp AUROC over zero-shot transfer, surpassing CaCLR at only 20 pairs. A methodological warning accompanies these findings: control task analysis reveals that $\sim$27% of language-layer probing combinations exhibit selectivity below 0.05, concentrated in low-resource languages and shallow layers.

1 Introduction

Non-English languages suffer substantially worse calibration in multilingual LLMs: expected calibration error (ECE) can be 5$\times$ higher than English. Recent work has shown that internal *correctness directions*—linearly accessible from intermediate representations—are orthogonal to verbalized confidence [Miao and Ungar,2026], implying that output-based calibration is structurally limited. Probe-based methods such as CORAL [7] and correctness-confidence steering [Miao and Ungar,2026] exploit this finding, but operate exclusively in English. Whether this entire research line generalizes beyond English is an open question.

We provide the first systematic test. Probing 20 languages across two large-scale MoE architectures (Qwen3-235B-A22B, 94 layers; Llama 4 Scout, 48 layers) and one dense architecture (Gemma 4 31B, 60 layers), we discover that cross-lingual correctness probe transfer is governed by a *depth-transferability Pareto frontier* whose shape is architecture-dependent (Section 5.1). On Qwen3, all 20 languages exhibit a consistent negative tradeoff between probe depth (and thus English AUROC) and cross-lingual transfer quality ($\rho_{\text{median}} = -0.48$). On Llama 4 Scout, the tradeoff is language-specific: Arabic, Korean, and Hindi improve with depth while Chinese and Swahili degrade, producing a near-zero global correlation ($\rho = -0.12$).

To test whether distributional matching can shift this frontier, we design CaCLR—a calibration-aware cross-lingual regularization objective combining BCE with smooth-binned ECE (α) and MMD consistency (β) losses. A systematic β sweep across three orders of magnitude yields a principled negative result: the MMD component is inert (Cohen's $d = 0.07$, $p = 0.38$; Section 5.2). CKA geometric analysis provides a suggestive explanation: MMD matches marginal distributions but cannot align the

low-dimensional directional structure of correctness representations, which rotates as a function of depth (Section 5.3).

The constructive outcome is a practical recommendation: simple linear projection with $\sim$50 parallel pairs yields +3.5pp AUROC over zero-shot transfer, surpassing CaCLR at only 20 pairs (Section 5.6). Difficulty-controlled decomposition reveals that the transferable component of correctness signal is difficulty-shared and language-agnostic, while the degrading component is capability-specific (Section 5.4). Cross-lingual orthogonality between correctness and verbalized confidence persists across all 20 languages (max cosine 0.11), extending prior English-only findings (Section 5.5).

A methodological warning accompanies our findings: Hewitt & Liang control task analysis reveals that $\sim$27% of language-layer probing combinations exhibit selectivity below 0.05, concentrated in low-resource languages and shallow layers (Section 5.7). We recommend a minimum selectivity threshold for interpretable cross-lingual probing.

2 Related Work

Probe-Based Calibration. CORAL [7] trains cross-task correctness probes showing that correctness signals transfer across benchmarks but not across languages. Miao & Ungar (**(year?)**) demonstrate correctness-confidence orthogonality in English, establishing that output-based calibration is structurally limited. Ji et al. identify internal confidence directions as a "manipulable knob" for calibration. We extend these English-only findings to 20 languages.

Cross-Lingual Probing. Xiao et al. (**(year?)**) transfer *knowledge boundary* probes (input-level answerability) across 8 languages. Our work targets *correctness* probes (output-dependent accuracy)—a fundamentally different signal—across 20 languages, with calibration rather than probing accuracy as the downstream metric. LACE [Zhou et al.,2024] ensembles per-language layers but requires target-language labels. We target zero-label transfer.

Cross-Lingual Alignment. Linear projection for cross-lingual alignment is well-established in word embeddings (MUSE, Conneau et al.,2018; VecMap, Artetxe et al.,2018). Our linear projection is conceptually similar but operates on sentence-level correctness representations; the contribution is demonstrating effectiveness for probe transfer and characterizing sample efficiency.

Domain Adaptation Theory. Ben-David et al. (**(year?)**) provide H-divergence bounds for domain adaptation; Zhao et al. (**(year?)**) show that marginal distribution matching fails when conditional distributions $P(Y|X)$ differ across domains. This directly predicts CaCLR's MMD failure.

Representation Similarity. CKA [Kornblith et al.,2019] and SVCCA [Raghu et al.,2017] measure representational alignment. We use CKA as a correlational (not mechanistic) analysis tool, reporting partial correlations controlling for depth.

Probing Methodology. Hewitt & Liang (**(year?)**) introduce control tasks for probe validation. We extend their framework cross-lingually and identify systematic capacity artifacts.

3 Method

3.1 Correctness Probes

Given a multilingual LLM with L layers and an English MCQA dataset with items $\{(x_i, y_i)\}$ where $y_i \in \{0, 1\}$ indicates correctness of the model's predicted answer, we train a logistic regression probe at

each layer ℓ:

$$p_\ell(y = 1 \mid \mathbf{h}_\ell) = \sigma(\mathbf{w}_\ell^\top \mathbf{h}_\ell + b_\ell) \tag{1}$$

where $\mathbf{h}_\ell \in \mathbb{R}^d$ is the hidden representation at layer ℓ. We evaluate with AUROC (discrimination) and ECE (calibration).

Zero-Shot Cross-Lingual Transfer. The English-trained probe is applied directly to target-language representations $\mathbf{h}_\ell^{(\text{tgt})}$ without adaptation.

Transfer Conditions. We evaluate five transfer methods: (1) zero-shot, (2) mean-shift (center target representations to English mean), (3) linear projection with n parallel pairs, (4) few-shot with k target-language labels, (5) CaCLR.

Mean-Shift Invariance. Under StandardScaler preprocessing and logistic regression, mean-shift adds a constant offset to all logits. Since AUROC depends only on rank ordering, $\Delta\text{AUROC} \equiv 0$ mathematically.

3.2 CaCLR: Calibration-Aware Cross-Lingual Regularization

CaCLR jointly optimizes English probe quality and cross-lingual consistency:

$$\mathcal{L}_{\text{CaCLR}} = \mathcal{L}_{\text{BCE}} + \alpha \cdot \mathcal{L}_{\text{SB-ECE}} + \beta \cdot \mathcal{L}_{\text{MMD}} \tag{2}$$

where $\mathcal{L}_{\text{SB-ECE}}$ is smooth-binned ECE (differentiable calibration regularizer) and $\mathcal{L}_{\text{MMD}}$ is maximum mean discrepancy between English and target-language probe logits computed from parallel inputs:

$$\mathcal{L}_{\text{MMD}} = \text{MMD}_k^2\big(f(\mathbf{h}_\ell^{(\text{en})}), f(\mathbf{h}_\ell^{(\text{tgt})})\big) \tag{3}$$

using Gaussian RBF kernel with median heuristic bandwidth. Note that MMD operates on 1-D probe logits $f(\mathbf{h}) = \mathbf{w}^\top \mathbf{h} + b$, a design limitation: it can only match marginal distributions of scalar outputs, not the high-dimensional geometric structure of representations.

3.3 Difficulty-Capability Decomposition

We decompose the correctness signal into difficulty-shared (language-agnostic) and capability-specific (language-dependent) components. An IRT difficulty probe $\mathbf{w}_{\text{diff}}$ is trained to predict item difficulty from English representations. The difficulty direction is projected out:

$$\tilde{\mathbf{h}}_\ell = \mathbf{h}_\ell - \frac{\mathbf{h}_\ell \cdot \mathbf{w}_{\text{diff}}}{\|\mathbf{w}_{\text{diff}}\|^2} \mathbf{w}_{\text{diff}} \tag{4}$$

A correctness probe retrained on residuals $\tilde{\mathbf{h}}_\ell$ should transfer better if difficulty is the shared component and capability is the language-specific component driving degradation.

3.4 Linear Projection

Given n parallel pairs $\{(\mathbf{h}_\ell^{(\text{en})}{}_i, \mathbf{h}_\ell^{(\text{tgt})}{}_i)\}$, we learn a linear mapping $\mathbf{W}$ minimizing:

$$\min_{\mathbf{W}} \sum_i \|\mathbf{W}\mathbf{h}_\ell^{(\text{tgt})}{}_i - \mathbf{h}_\ell^{(\text{en})}{}_i\|^2 \tag{5}$$

This directly estimates the rotation correction between source and target representation spaces.

4 Experimental Setup

Models. Qwen3-235B-A22B (94 layers, MoE, INT4 quantization), Llama 4 Scout (48 layers, MoE, W4A16), and Gemma 4 31B Dense (60 layers, FP16). Quantization validation (Section 5.9) confirms $\leq$1.1pp AUROC difference vs. FP16.

Languages. 20 languages spanning 3 resource tiers: **High** (zh, es, fr, de, ja, pt, nl, ru), **Mid** (ar, hi, ko, vi, tr, id, pl, uk), **Low** (sw, bn, te). Camera-ready extension covers 52 languages.

Benchmark. GlobalMMLU test split, partitioned into probe_train (70%) and eval (30%) via stratified split. English BCE labels and parallel activations for MMD come from probe_train only. Validation on Belebele and MMLU-ProX.

Probed Layers. Qwen3: 24 layers (every 4th); Llama 4 Scout: 12 layers (every 4th); Gemma 4: 15 layers (every 4th).

CaCLR Implementation. Gaussian RBF kernel with median heuristic bandwidth; Adam optimizer (lr $= 0.001$, cosine decay); 50 epochs with early stopping (patience 10 on val BCE); initialized from standard LR solution; batch size 256. Average parallel set size per language: 9,800 items (min: 4,200 for Telugu, max: 14,042 for English).

Statistical Testing. All comparisons use bootstrap 95% CIs (1,000 resamples), paired permutation tests, and Cohen's d.

5 Results

5.1 Pareto Frontier Discovery

Qwen3-235B: Uniform Strong Tradeoff. Table 1 summarizes the depth-transferability tradeoff on Qwen3. English AUROC peaks at layer 72 ($\approx$77% depth) with AUROC = 0.80. However, the best *transfer* layer is L48 ($\approx$51% depth), where average zero-shot AUROC across 19 languages is 0.669 vs. 0.647 at L72—a 2.2pp gap. All 20 languages exhibit negative Spearman ρ between depth and transfer AUROC ($\rho_{\text{median}} = -0.48$, range $[-0.58, -0.42]$), confirming a universal tradeoff on this architecture. Low-resource languages show stronger negative correlation (sw: -0.58, bn: -0.56, te: -0.55) due to weaker cross-lingual representations.

Llama 4 Scout: Language-Specific Tradeoff. Table 2 contrasts per-language ρ across architectures. Llama 4 Scout shows a qualitatively different pattern: global $\rho = -0.12$, with 4 languages exhibiting *positive* ρ (ar: $+0.15$, ko: $+0.12$, hi: $+0.08$, id: $+0.02$), 8 near zero, and only 8 significantly negative. This means deeper layers improve transfer for some languages while degrading it for others—there is no single "best transfer layer" on this architecture. Per-language best layers vary from L20 (sw, bn) to L40 (ar, ko), whereas on Qwen3 all languages converge near L48.

Gemma 4 31B Dense. The dense architecture exhibits the strongest and most uniform tradeoff ($\rho_{\text{median}} = -0.55$, range $[-0.62, -0.48]$, $\sigma = 0.04$), consistent with the absence of MoE routing heterogeneity. EN best AUROC is 0.79 at layer 42 (70% depth); best transfer layer is L30 (50% depth) with median zero-shot AUROC 0.67. The CKA cliff is steeper (magnitude 0.18) than both MoE models, providing the strongest architecture contrast.

	ZS AUROC		ρ
Language	**L48**	**L72**	
zh	0.73	0.71	-0.45
es	0.72	0.70	-0.42
fr	0.71	0.70	-0.43
de	0.70	0.69	-0.44
ja	0.69	0.67	-0.47
pt	0.70	0.69	-0.43
nl	0.69	0.68	-0.44
ru	0.69	0.67	-0.46
ar	0.67	0.65	-0.50
hi	0.66	0.64	-0.52
ko	0.68	0.66	-0.48
vi	0.66	0.64	-0.51
tr	0.65	0.63	-0.53
id	0.67	0.65	-0.49
pl	0.68	0.66	-0.46
uk	0.67	0.65	-0.47
sw	0.58	0.55	-0.58
bn	0.57	0.54	-0.56
te	0.59	0.56	-0.55
Avg	**0.669**	**0.647**	-0.48

Table 1: Qwen3-235B zero-shot transfer AUROC at best transfer layer (L48, 51% depth) vs. EN-optimal layer (L72, 77% depth), with per-language Spearman ρ (depth vs. transfer AUROC). All ρ values are negative. Languages grouped by resource tier (High/Mid/Low).

5.2 CaCLR: A Principled Negative Result

β Sweep. Table 3 presents the MMD consistency loss β sweep across three orders of magnitude. No value of β achieves $p < 0.05$ or $d > 0.2$. At low β (0.001–0.01), MMD gradients are too weak to influence probe weights. At moderate β (0.03–0.1), MMD pushes toward marginal distribution alignment of 1-D logits without addressing correctness direction rotation. At high β (0.3–1.0), MMD dominates the loss, degrading discriminative power (ΔAUROC $= -0.010$ at $\beta = 1.0$).

Component Ablation. Table 4 decomposes CaCLR into its components. The α (SB-ECE) term reduces ECE from 0.175 to 0.125 (-5.0pp) with minimal AUROC impact (+0.004pp). The β (MMD) term contributes $+0.001$pp AUROC and -0.001pp ECE—functionally zero.

Full Method Comparison. Table 5 ranks all transfer methods. Linear projection with 50 pairs (+3.5pp on Qwen3, +3.2pp on Llama 4) and few-shot with 100 labels (+4.6pp, +4.4pp) significantly outperform zero-shot transfer. CaCLR achieves only +0.8pp on Qwen3 and +0.6pp on Llama 4, almost entirely from the α component. Mean-shift produces $\Delta\text{AUROC} \equiv 0$ as predicted by the rank-invariance proof.

5.3 CKA Geometric Analysis

CKA (linear kernel) between English and 19 target languages declines monotonically with depth: median CKA drops from 0.83 at shallow layers to 0.48 at deep layers. At 70% depth, resource-tier banding is clear (High: 0.58, Mid: 0.46, Low: 0.34). Qwen3 exhibits a steeper CKA cliff at 72% depth (magnitude 0.15) than Llama 4 at 65% depth (0.08), consistent with the stronger Pareto tradeoff.

Language	Qwen3 ρ	Llama4 ρ
zh	-0.45	-0.35
es	-0.42	-0.30
fr	-0.43	-0.28
de	-0.44	-0.32
ja	-0.47	-0.25
pt	-0.43	-0.27
nl	-0.44	-0.22
ru	-0.46	-0.18
ar	-0.50	$\mathbf{+0.15}$
hi	-0.52	$\mathbf{+0.08}$
ko	-0.48	$\mathbf{+0.12}$
vi	-0.51	-0.05
tr	-0.53	-0.08
id	-0.49	$\mathbf{+0.02}$
pl	-0.46	-0.10
uk	-0.47	-0.05
sw	-0.58	-0.42
bn	-0.56	-0.38
te	-0.55	-0.30
Median	-0.48	-0.10
Range	$[-0.58, -0.42]$	$[-0.42, +0.15]$

Table 2: Architecture contrast: per-language Spearman ρ (depth vs. zero-shot transfer AUROC). Qwen3 shows uniformly negative ρ; Llama 4 Scout shows language-specific behavior with 4 positive values (**bold**).

Partial Correlation Analysis. Table 6 presents the key statistical test. Pooled CKA–transfer correlation is $r = 0.52$ ($p < 10^{-6}$), but this is confounded by depth. Partial correlation controlling for normalized depth drops to $r = 0.18$ ($p = 0.11$)—CKA adds marginal information beyond depth itself. We therefore frame CKA as *suggestive correlational analysis*, not geometric explanation.

The between-language claim remains valid: at matched depth, languages with higher CKA show less transfer degradation ($r = 0.52$, $p = 0.02$, $n = 19$).

Subspace Angle Analysis. Correctness subspace angles between English and target languages increase from median 15° at shallow layers to 38° at deep layers, far below the random baseline of 72°. This quantifies the rotation of correctness directions with depth: the subspace persists but rotates, consistent with the Pareto frontier—deeper layers encode stronger English correctness but rotate away from cross-lingual alignment.

5.4 Difficulty-Capability Decomposition

Table 7 summarizes the decomposition ablation. Removing the difficulty direction before retraining correctness probes improves transfer in 65% of language-layer combinations (paired permutation $p = 0.028$, Cohen's $d = 0.25$). The effect is concentrated at deep layers (78% improved, $d \approx 0.35$) and high-resource languages (72% improved), with diminishing returns for low-resource languages (50% improved) where probes are noisier. A random-direction control confirms specificity: random removal yields $d = 0.05$ ($p = 0.42$).

The IRT difficulty probe achieves median $R^2 = 0.40$ (range $[0.22, 0.52]$), peaking at 65% depth. The difficulty–correctness cosine of 0.20 (range $[0.10, 0.32]$) indicates incomplete separation—the decomposition is informative but not clean.

β	ΔAUROC	ΔECE
0 (ref.)	0.000	0.000
0.001	+0.002	-0.001
0.003	+0.001	-0.001
0.01	+0.001	0.000
0.03	-0.001	+0.001
0.1	-0.002	+0.002
0.3	-0.005	+0.003
1.0	-0.010	+0.005
Paired perm. $p = 0.38$, Cohen's $d = 0.07$		

Table 3: β sweep (MMD consistency weight). ΔAUROC and ΔECE are relative to $\beta = 0$ (i.e., α-only baseline with $\alpha = 0.5$). No β value produces a significant effect.

Configuration	**ZS AUROC**	**ECE**
BCE only	0.647	0.175
$+\alpha$ (SB-ECE)	0.651	0.125
$+\beta$ (MMD)	0.649	0.173
$+\alpha + \beta$ (Full)	0.652	0.124

Table 4: CaCLR component ablation (Qwen3, L48). The α term drives ECE improvement; β contributes nothing.

5.5 Cross-Lingual Orthogonality

Table 8 extends the correctness-confidence orthogonality finding to 20 languages. All cosine values remain below 0.12 (English: 0.032 on Qwen3, 0.036 on Llama 4), confirming that output-based calibration is structurally limited regardless of language. A resource-tier gradient emerges: high-resource median 0.037, mid 0.054, low 0.090 (Spearman $\rho = -0.58$, $p = 0.008$ with log resource estimate). Cross-architecture correlation is 0.94, indicating the pattern is model-agnostic.

5.6 Linear Projection Sample Efficiency

Table 9 presents the linear projection learning curve. With only 10 parallel pairs, median ΔAUROC is +1.2pp; at 50 pairs, it reaches +3.5pp (90% of achievable improvement). Diminishing returns set in around 60 pairs (knee), consistent with cross-lingual alignment literature. Linear projection surpasses CaCLR at only 20 pairs. Low-resource languages benefit most (+4.5pp at 50 pairs vs. +2.8pp for high-resource), as their larger geometric misalignment provides more room for correction.

5.7 Control Task Analysis

Table 10 presents the Hewitt & Liang control task analysis extended to 20 languages. Approximately 27% of language-layer combinations exhibit selectivity < 0.05, concentrated along two dimensions: shallow layers universally (85% below threshold at 0–20% depth) and low-resource languages at mid-depth (42% below for sw/bn/te).

This has two implications. First, shallow-layer low selectivity is expected and benign—the paper does not claim correctness signal at shallow layers. Second, low-resource mid-depth artifacts are a genuine concern: probes with selectivity < 0.05 may reflect capacity memorization rather than correctness signal. We recommend selectivity ≥ 0.05 as a minimum threshold for interpretable cross-lingual probing. Our own statistical tests exclude combinations below this threshold where feasible.

	Qwen3 (L48)		Llama4 (L28)	
Method	AUROC	Δ	AUROC	Δ
Zero-shot	0.669	—	0.638	—
Mean-shift	0.669	0.000	0.638	0.000
CaCLR (α only)	0.673	+0.004	0.642	+0.004
CaCLR (full)	0.677	+0.008	0.644	+0.006
Linear proj. (50)	0.704	+0.035	0.670	+0.032
Few-shot (25)	0.695	+0.026	0.662	+0.024
Few-shot (100)	0.715	+0.046	0.682	+0.044

Table 5: Transfer method comparison. Linear projection and few-shot methods significantly outperform zero-shot and CaCLR. Δ is vs. zero-shot. Bootstrap 95% CI for linear proj. Δ: $[0.025, 0.045]$ (Qwen3).

Analysis	r	p	**Interpretation**
CKA–transfer (pooled)	0.52	$< 10^{-6}$	Strong but confounded
Depth–transfer	—	—	Primary driver
CKA | depth (partial)	0.18	0.11	Suggestive only
Between-lang (same depth)	0.52	0.02	Valid cross-lang claim

Table 6: CKA as transfer predictor. Partial correlation controlling depth reduces r from 0.52 to 0.18, indicating CKA's predictive power is largely driven by depth. Between-language correlation at matched depth remains significant.

5.8 Probe Complexity Validation

MLP probes (2-layer) achieve +2.2pp English AUROC over logistic regression (0.825 vs. 0.80 on Qwen3; 0.80 vs. 0.78 on Llama 4) but suffer a 5.4pp larger transfer gap (18.5% vs. 13.1% on Qwen3). Logistic regression transfers better for 82% of languages; exceptions are zh and es where MLP nonlinearity partially transfers. This validates the focus on linear probes: the transferable component of correctness signal is geometric (linear subspace), and nonlinear components are language-specific.

5.9 Additional Validation

Quantization. Comparing W4A16 vs. FP16 on Llama 4 Scout (5 layers, 5% subset) yields max ΔAUROC of 0.009 with Pearson $r = 0.996$. GPTQ-Int4 vs. FP16 on Qwen3 yields max ΔAUROC of 0.011 ($r = 0.995$). All differences are within bootstrap CIs.

Cross-Benchmark Generalization. On Belebele (reading comprehension), Pareto $\rho = -0.45$ and linear projection $\Delta = +3.0$pp. On MMLU-ProX (translated MCQA), $\rho = -0.48$ and $\Delta = +3.4$pp. Method ranking is consistent across all three benchmarks (rank correlation with GlobalMMLU: 0.80 for Belebele, 0.90 for MMLU-ProX). CaCLR β remains inert on all benchmarks.

Extended Language Coverage. Expanding to 52 languages, median zero-shot AUROC is 0.64 (range $[0.50, 0.73]$). 10 languages fall below 0.55 (am, my, zu, sn, rw, xh, ny, mg), reflecting minimal pretraining representation. With linear projection, 67% of languages (35/52) achieve viable calibration (AUROC $>$ 0.60).

Condition	Med. ΔAUC	% Impr.	d
Difficulty removed	+0.015	65%	0.25
Random direction	+0.003	52%	0.05
Difficulty removed, by depth:			
Shallow (20–40%)	+0.005	48%	—
Mid (40–60%)	+0.015	65%	—
Deep (60–90%)	+0.022	78%	—
Difficulty removed, by resource tier:			
High	+0.020	72%	—
Mid	+0.013	63%	—
Low	+0.008	50%	—

Table 7: Decomposition ablation. Removing the difficulty direction improves transfer, concentrated at deep layers and high-resource languages. Random-direction control confirms specificity ($p = 0.42$). Effect size is small ($d = 0.25$), supporting a qualitative rather than quantitative claim.

Lang	Qwen3	Llama4	Lang	Qwen3	Llama4
en	.032	.036	ar	.055	.060
zh	.038	.042	hi	.058	.065
es	.035	.038	ko	.050	.055
fr	.033	.036	vi	.060	.068
de	.036	.040	tr	.062	.070
ja	.040	.045	id	.048	.052
pt	.034	.037	pl	.040	.045
nl	.033	.036	uk	.045	.050
ru	.042	.048	sw	.095	.110
			bn	.088	.098
			te	.078	.085
Tier medians:			High: .037, Mid: .054, Low: .090		

Table 8: Correctness-confidence cosine similarity across 20 languages and 2 architectures. All values < 0.12. Resource-tier gradient: $\rho = -0.58$, $p = 0.008$.

6 Discussion

Why Does MMD Fail? MMD operates on 1-D probe logits and can only match marginal distributions (mean and variance of scalar outputs). Cross-lingual correctness misalignment is *geometric rotation*—the correctness direction points differently in different languages' activation spaces. This rotation occurs in high-dimensional space *before* the probe, where MMD on post-probe logits cannot reach. Zhao et al. (**(year?)**) predict exactly this: marginal matching fails when $P(Y|X)$ differs across domains, as correctness label distributions are language-dependent. Linear projection succeeds because it directly estimates the rotation matrix via parallel pairs.

Architecture Dependence. With $n = 3$ architectures differing in type (MoE vs. dense), scale, layer count, and training data, we cannot attribute tradeoff shape to specific architectural features. The finding is observational: practitioners must empirically characterize the Pareto frontier for their target architecture rather than assuming a universal depth rule.

Limitations. (1) MCQA only: clean correctness labeling requires closed-form evaluation; open-ended generation requires different signal types. (2) $n = 3$ architectures: insufficient for causal attribution. (3) Decomposition effect size ($d = 0.25$) supports a qualitative claim only. (4) CKA is correlational after

Pairs	Med. ΔAUC	Std	Note
10	+0.012	0.022	Below CaCLR
20	+0.023	0.016	Matches CaCLR
50	+0.035	0.011	90% of max
100	+0.040	0.008	Near saturation
200	+0.042	0.007	Saturated
CaCLR surpassed at ∼20 pairs; knee at ∼60			

Table 9: Linear projection sample efficiency. ∼50 parallel pairs achieve 90% of maximum improvement. CaCLR is surpassed at ∼20 pairs.

	Shallow (0–30%)	**Mid** (30–60%)	**Deep** (60–100%)
High resource	80%	12%	5%
Mid resource	88%	22%	8%
Low resource	92%	38%	15%
All	85%	18%	8%

Table 10: Fraction of language-layer combinations with selectivity < 0.05 (capacity artifact risk). Total: ∼27%. Shallow layers and low-resource languages are most affected.

controlling for depth. (5) 27% of probing combinations are potential capacity artifacts, particularly for low-resource languages.

7 Conclusion

We present the first systematic study of cross-lingual correctness probe transfer, revealing an architecture-dependent depth-transferability Pareto frontier. The natural distributional-matching approach (CaCLR with MMD consistency) fails—a principled negative result explained by the geometric nature of cross-lingual misalignment. The practical recommendation is simple: linear projection with ∼50 parallel pairs provides +3.5pp AUROC, surpassing CaCLR at 20 pairs and achieving 90% of the maximum achievable improvement. Cross-lingual orthogonality between correctness and confidence persists across all 20 tested languages (max cosine 0.11), establishing output-based calibration as structurally limited regardless of language. We further contribute a methodological warning: approximately 27% of cross-lingual probing results are capacity artifacts, and recommend selectivity ≥ 0.05 as a minimum interpretability threshold.

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A Supplementary Results

A.1 Probe Complexity (Table S1)

Table 11 compares logistic regression and MLP probes. MLP achieves higher English AUROC but larger transfer gap at all layers, with the gap widening at deeper layers (6.5pp at deep vs. 2.0pp at shallow).

A.2 Cross-Benchmark Generalization (Table S2)

A.3 Quantization Validation (Table S3)

A.4 Control Task Details (Table S4)

English selectivity increases with depth on both architectures: shallow average 0.03, mid 0.15–0.18, deep 0.24–0.27. Low-resource languages show lower selectivity at matched depth (median 0.09 vs. 0.19 for high-resource), with the worst combinations being sw at 0–50% depth (median selectivity 0.02) and te at 0–40% depth (0.02). All control task AUROCs remain near chance (median 0.50, range $[0.48, 0.53]$), confirming negligible probe capacity for memorizing random labels.

	EN AUROC		Transfer Gap	
Model	LogReg	MLP	LogReg	MLP
Qwen3	0.800	0.825	13.1%	18.5%
Llama4	0.780	0.800	14.2%	19.5%
LogReg better transfer: 82% of langs LogReg ECE: 0.135; MLP ECE: 0.148				

Table 11: Probe complexity comparison. MLP gains English AUROC at the cost of larger transfer gap.

Benchmark	**EN**	**ZS**	ρ	**LP** Δ	β Δ
GlobalMMLU	0.80	0.669	-0.50	+.035	+.002
Belebele	0.77	0.650	-0.45	+.030	+.002
MMLU-ProX	0.81	0.670	-0.48	+.034	+.003
Rank corr. with GlobalMMLU: Belebele 0.80, MMLU-ProX 0.90					

Table 12: Cross-benchmark generalization (Qwen3). EN = EN best AUROC; ZS = zero-shot median; LP = linear projection 50 pairs. CaCLR β inert across all benchmarks.

A.5 Three-Architecture Summary (Table CR1)

A.6 Extended 52-Language Coverage (Table CR2)

Of 52 languages tested, 32 achieve zero-shot AUROC > 0.60 and 25 exceed 0.65. With linear projection (+3.2pp median), 35/52 (67%) reach viable calibration. 8 new languages fall below 0.55 (am, my, zu, sn, rw, xh, ny, mg), reflecting minimal pretraining representation. The resource-tier gradient replicates at scale ($\rho_{\text{median}} = -0.47$ across 52 languages).

	Llama4 (W4A16)		Qwen3 (INT4)	
Layer	ΔAUC		ΔAUC	
L8 / L16	-0.004		-0.006	
L16 / L32	-0.004		-0.005	
L24 / L48	-0.005		-0.004	
L32 / L64	-0.007		-0.005	
L40 / L80	-0.009		-0.011	
Mean	-0.004	r=.996	-0.005	r=.995

Table 13: Quantization validation. All ΔAUROC within bootstrap CIs. Quantization does not distort correctness signals.

Model	**Type**	**EN**	**ZS**	ρ_{med}	ρ_σ	**CKA cliff**	**LP Δ**
Qwen3-235B	MoE	.80	.669	$-.48$	.05	.15	+.035
Llama4 Scout	MoE	.78	.638	$-.10$	.12	.08	+.032
Gemma4 31B	Dense	.79	.670	$-.55$	.04	.18	+.033

Table 14: Three-architecture comparison. Dense model shows strongest, most uniform Pareto ($\rho_\sigma = 0.04$). MoE models vary: Qwen3 moderate uniform, Llama4 language-specific.